EMBEDDED SYSTEM AND DESIGN
EXPERIMENT NO. 3

CLOCK

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Course/Section: CPE160P-4/A1

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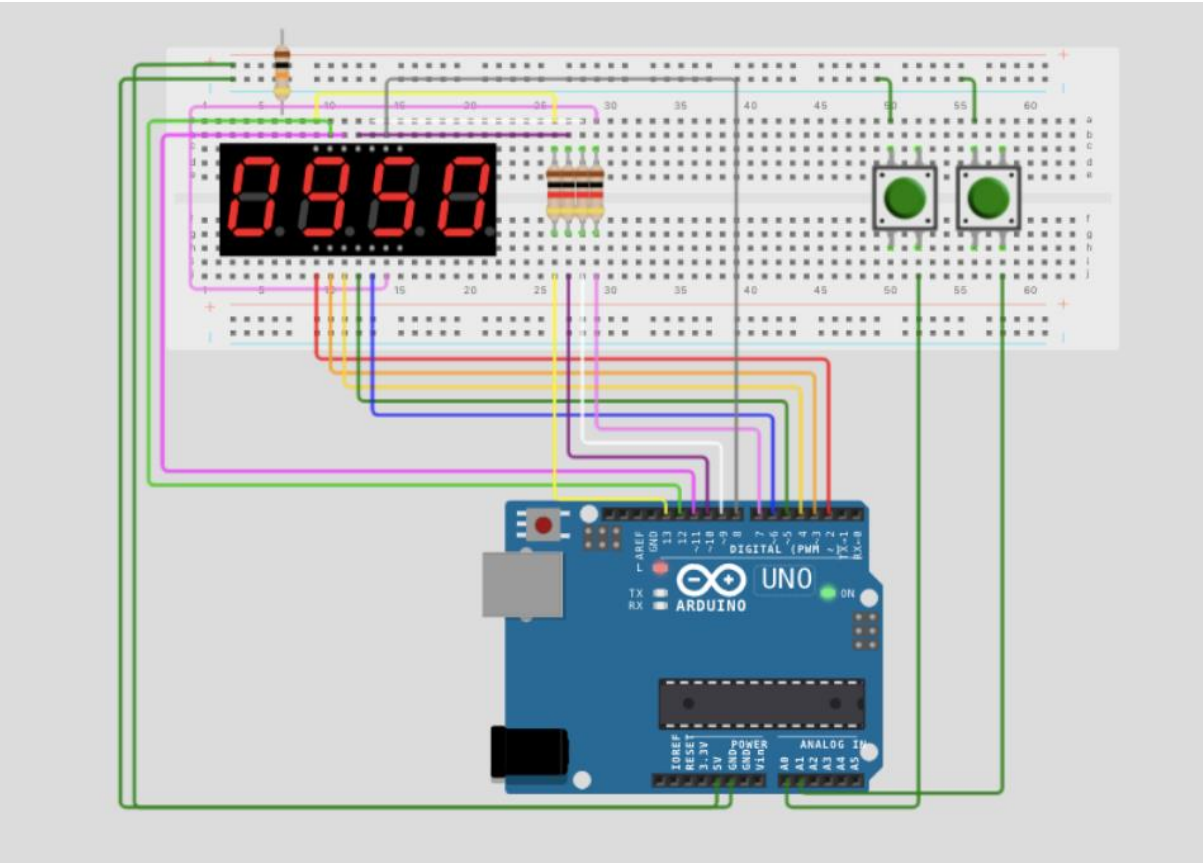
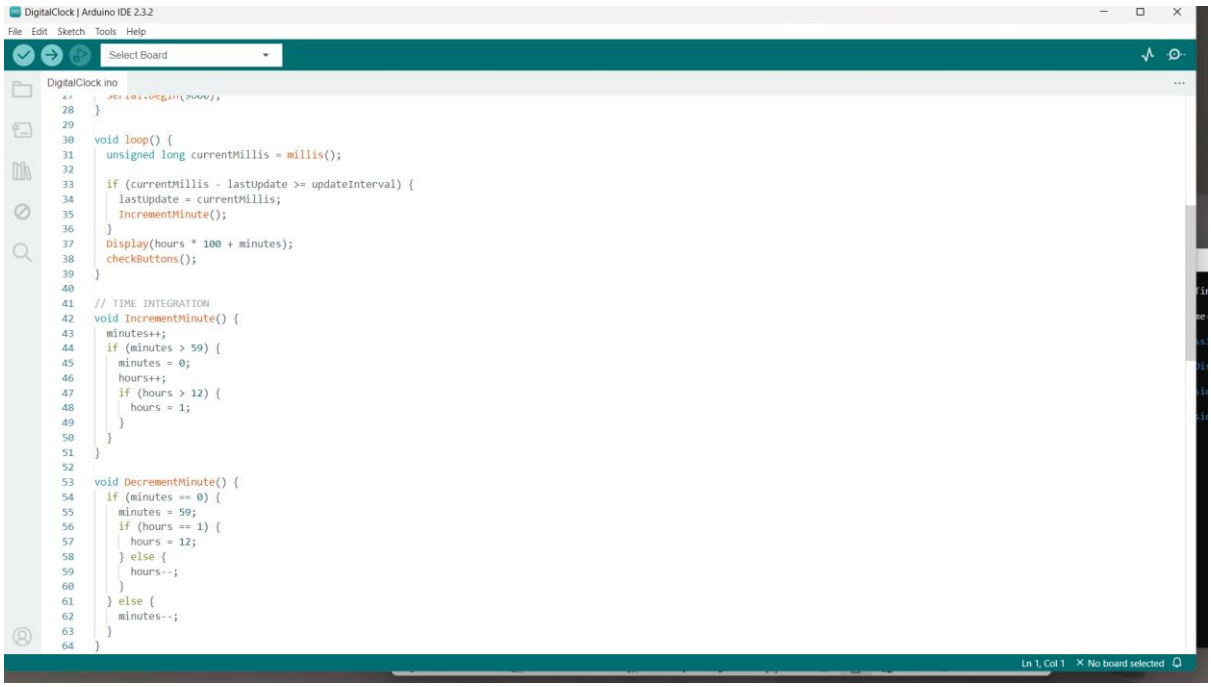
DISCUSSION

Experiment 3 titled Clock revolves around the idea of creating a digital working clock using 3 main components namely, 2 buttons, 1 4 digit 7 segments, and Arduino board. The first button was designed to increment the minutes by 1 with a short press or increment the hours by 1 when held for more than 1 second. The second have the same concept as the first button yet it decrements the digit to 1minutes or 1hour if held for more than 1 second.



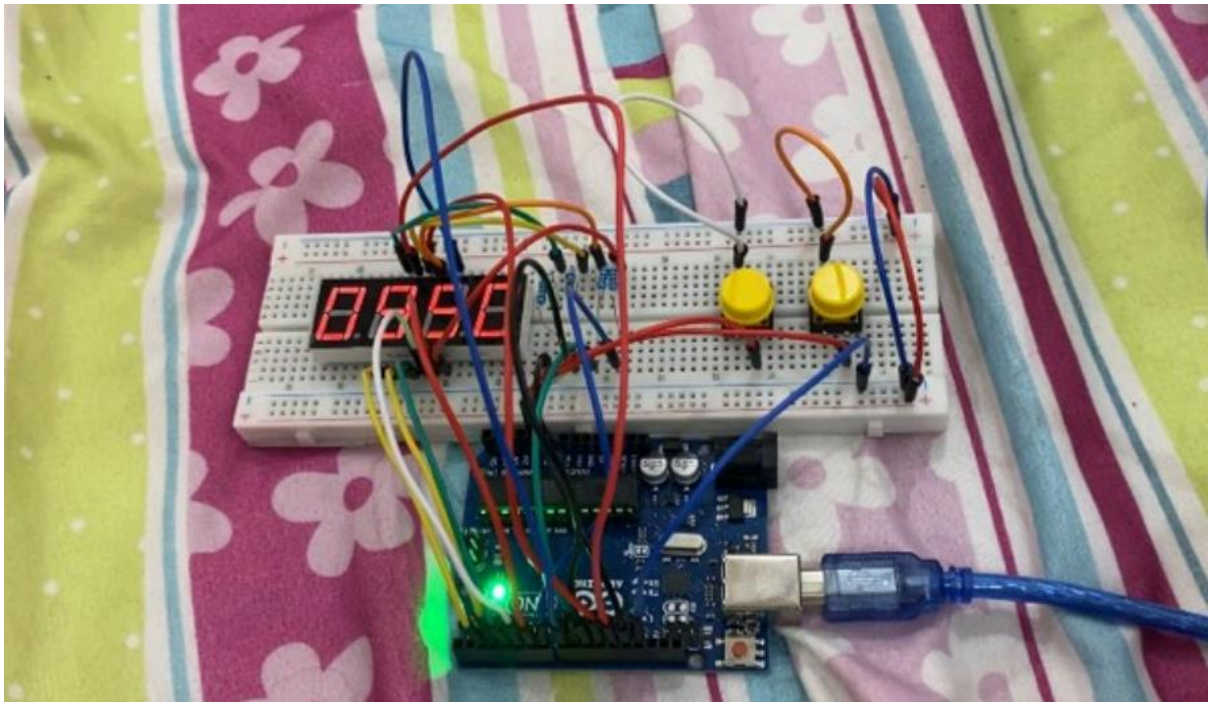
The first step to satisfy the experiment main objective is to prepare and simulate the desired circuit in a simulator. I used wikwo rather than tinker cad for this part as tinker cad still doesn't have a 4-digit 7 segment in its component. For this part handling the 7 segment pins and determining if the segment i have is anode or cathode is a challenge. Not until the last part as I thought the 7 segment I have is an anode but

cathode instead, thus cause me trouble in the last part of the experiment, yet later fix in the end. After creating the circuit on the simulator, coding the timings and initializing all the button pin outs is essential.



The code consists of 5 main parts. Initializations of pins, Main function loop, Time Functions, Button Functions, and 7 segment functions. The initialization and main circuit consist of all the pin outs and the main function is what I want to

see in my circuit. Time functions consist of all the minutes and hour changing algorithms and will be use by the next part, the button functions which handles all the button related input and output I want to receive. The 7 segment functions handle all the output to be displayed to the 7 segment and its timing. It includes all 0 to 9 states of 7 segment to be used in each part of the 4 digit of the 7 segments.



After simulating and coding, physical implementation of circuit diagram is done together with uploading the code to the Arduino. As said the 7-segment display being configured as a common anode, while the code and hardware setup were intended for a common cathode configuration set some trouble. Yet, through thoughtful coding and iterative problem-solving, I was able to implement both minute and hour adjustments using simple button inputs.

CONCLUSION

The experiment objective is fully achieved through designing and integrating a digital clock with two buttons for incrementing and decrementing the initial value of clock. Using Arduino, we incorporated a basic mechanism how embedded system can be implemented with various functionality and useful real-life application. The detection of button if it s held or pressed is a simple fundamental concept, where devices communication and synchronize operation is achieved.

The use of 4-digit seven segment also brings us back to the its fundamental core that was a topic in Logic Circuit and Design. The circuit diagram on

how to connect 4 single seven segment was reviewed and how each pin in the 4-digit seven segment represents each other. The application of switching theory is shown in button handling and debouncing logic. These theoretical knowledge shows how we can apply it in practical applications.

The overall experience and concept of the experiment demonstrate the functionality of how a real clock works and how to apply embedded system to show hours and minutes. By programming the Arduino bored and integrating all circuit pins, we stabilized a control over our own digital clock. the experiment showcased the versatility of embedded systems in controlling

hardware and displaying information dynamically.

In Conclusion, the experiment showcase hands-on experience in coding and handling a 4 digit seven segment and buttons. With right coding of timing and updates we stabilize a right delay for the my digital clock. This gives me comprehensive understanding of clock coding and theoretical knowledge through practical application.